DIFFERENTIALS OF THE UNIVERSAL FIRST ORDER THICKENING RESTRICT TO THE DIFFERENTIALS OF THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1332-1342 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 1344-1357, the complete author proof (12 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Universal first order thickenings of formally unramified morphisms, conormal sheaves, and the transitivity sequence for differentials; the comparison is retained.

Source reference. Original lemma algebra-lemma-differentials-universal-thickening: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1357–1357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-universal-thickening-localize: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1310–1319. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-differentials-restrict-open: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1357–1357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma morphisms-lemma-functoriality-differentials: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 1357–1357. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $h : Z \to X$ be a formally unramified morphism of schemes over S. Let $Z \subset Z'$ be the universal first order thickening of Z over X with structure morphism $h' : Z' \to X$. The canonical map*

$$c_{h'} : (h')^*\Omega_{X/S} \longrightarrow \Omega_{Z'/S}$$

induces an isomorphism $h^*\Omega_{X/S} \to \Omega_{Z'/S} \otimes \mathcal{O}_Z$.

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. The map $c_{h'}$ is the map defined in Morphisms, Lemma morphisms-lemma-functoriality-differentials. If $i : Z \to Z'$ is the given closed immersion, then $i^*c_{h'}$ is a map $h^*\Omega_{X/S} \to \Omega_{Z'/S} \otimes \mathcal{O}_Z$. Checking that it is an isomorphism reduces to the affine case by localization, see Lemma lemma-universal-thickening-localize and Morphisms, Lemma morphisms-lemma-differentials-restrict-open. In this case the result is Algebra, Lemma algebra-lemma-differentials-universal-thickening. □